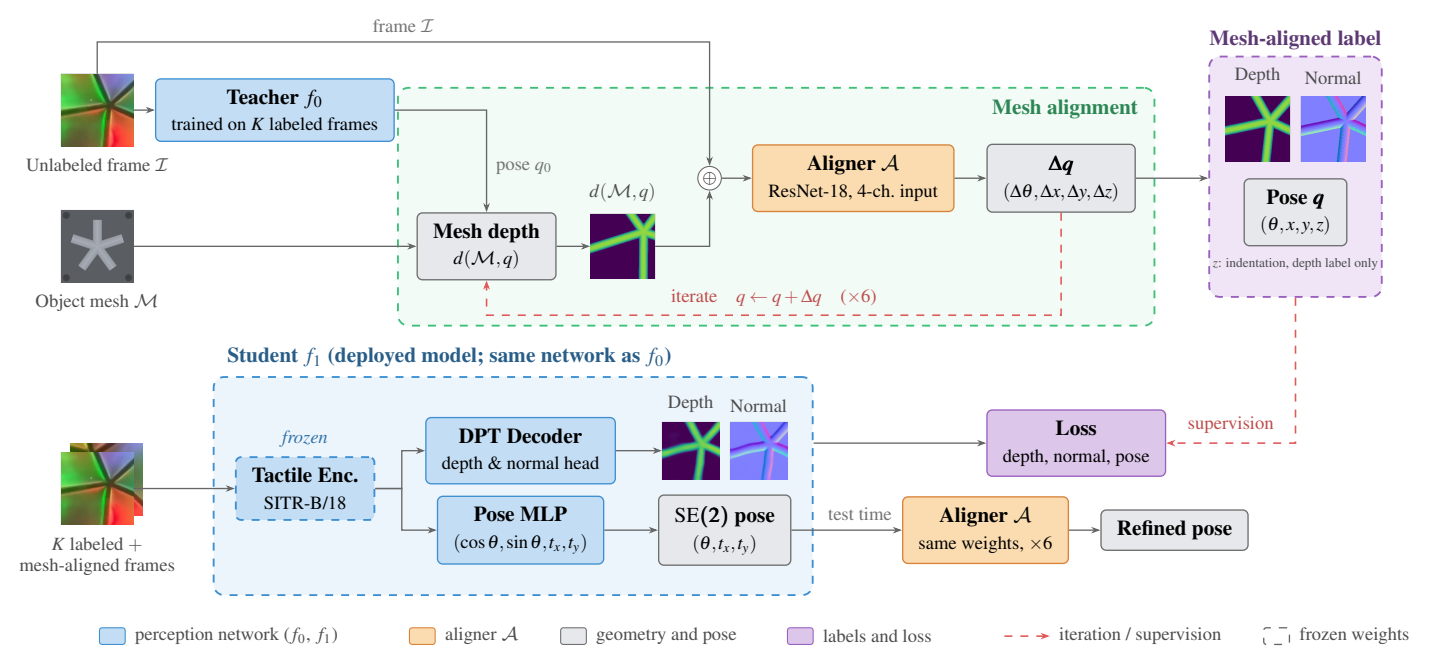

frame $\mathcal{I}$
Unlabeled frame $\mathcal{I}$
Teacher f_0
trained on K labeled frames
Object mesh $\mathcal{M}$
pose q_0
Mesh depth
$d(\mathcal{M}, q)$
$d(\mathcal{M}, q)$
Mesh alignment
Aligner $\mathcal{A}$
ResNet-18, 4-ch. input
Δq
$(\Delta\theta, \Delta x, \Delta y, \Delta z)$
iterate $q \leftarrow q + \Delta q$ $(\times 6)$
Mesh-aligned label
Depth
Normal
Pose q
(θ, x, y, z)
z: indentation, depth label only
Student f_1 (deployed model; same network as f_0)
K labeled + mesh-aligned frames
frozen
Tactile Enc.
SITR-B/18
DPT Decoder
depth & normal head
Pose MLP
$(\cos\theta, \sin\theta, t_x, t_y)$
Depth
Normal
SE(2) pose
(θ, t_x, t_y)
Loss
depth, normal, pose
supervision
test time
Aligner $\mathcal{A}$
same weights, ×6
Refined pose
perception network (f_0, f_1)
aligner $\mathcal{A}$
geometry and pose
labels and loss
iteration / supervision
frozen weights